

Business questions, answered with data

Statistics / Data and Business Analytics

Turning analysis into evidence-based decisions.

Statistics undergraduate | Expected graduation: July 2027

Data analytics, business analysis, and data-driven operations

SQL / Python / Statistics / Business Analysis

PENG
ZHUO

Evidence starts with a business question

Statistics background | Data, business, and operations analytics

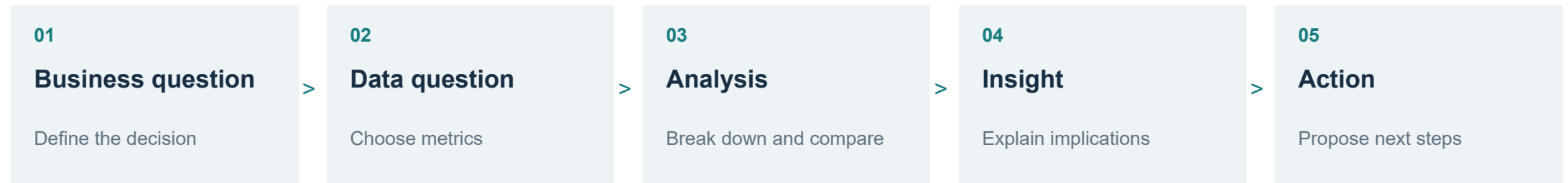
Background

Statistics undergraduate, graduating in July 2027.
I define business questions, examine data, and turn evidence into business judgments.

Analytical toolkit

SQL, Python, Pandas, and statistical analysis.
Projects cover metric design, result validation, survey research, and model development.

How I work



Focus: clear problem framing, metric definitions, credible evidence, and operational priorities.

Three contexts, one analytical approach

Understand the business, define metrics, and use evidence to guide decisions

01 E-commerce fulfillment and customer experience

Public-data operations analysis | SQL / Pandas

Question Where should the team investigate delays first?

Evidence The logistics stage has the largest time gap; RJ combines risk with material volume.

My contribution: metric design, SQL implementation, and result validation.

02 Digital transformation of a Luzhou seafood market

Market research project | Survey / K-means / OLS

Question Which digital services merit a pilot, and for whom?

Evidence Transparent weighing ranks first; three customer groups and a sales decline guide pilots.

My contribution: survey design, field research, and core data analysis.

03 Aging and health-system resilience in Chongqing

Statistical modeling competition | Entropy weighting / Panel models

Question How can capacity be measured to inform differentiated resource planning?

Evidence A four-dimension index; the aging-resilience association varies with local economic conditions.

My contribution: data collection, analysis, model building, and conclusions.

Turn fulfillment performance into decisions

Public order data used to simulate a service-operations investigation

01 Which stage?

Merchant handling / logistics

Metric: average stage duration

Method: on-time versus late

02 Which locations?

State / city / seller origin

Metrics: late rate, volume, severity

Method: aggregation and routes

03 What about ratings?

Delivery outcome / reviews

Metrics: average rating, low-score rate

Method: order-level comparison

Data

Orders, customers, order items, sellers, and reviews; the order is the unit of analysis.

My role

Metric design, SQL implementation, and result validation.
Python visuals are included as project outputs.

Timing

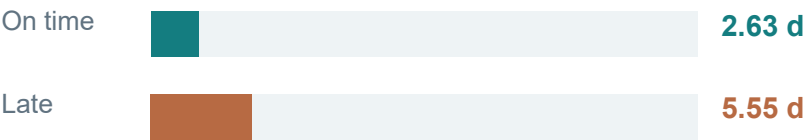
Completed September 1, 2026, using public orders from 2016-2018.

The largest time gap is in logistics

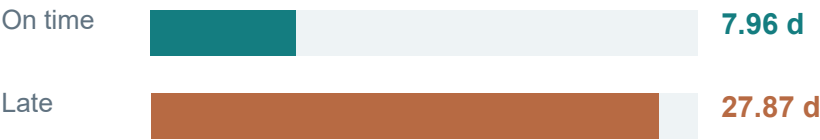
Stage comparison first; rate and volume then guide geographic priorities

Average duration by stage / days

Merchant handling



Logistics



Gap: 19.90 days in logistics versus 2.92 days in merchant handling.

Implication: inspect logistics first. Stage averages do not identify a carrier node or prove root cause.

State-level rate and volume

State	Orders	Late	Rate
SP	40,494	1,820	4.49%
RJ	12,350	1,495	12.11%
MA	717	125	17.43%

RJ: 12.11% late; 1,495 late orders.

RJ is not highest on either measure alone. Its combination of rate and volume supports priority review.

RJ needs distinct city and route responses

Volume, risk, and delay severity call for different operational checks

City priorities / existing findings

Volume

Rio de Janeiro

706 late orders; 47.22% of RJ late orders

Risk

São Gonçalo

83 of 385 orders late; 21.56% late rate

Severity

Nova Friburgo

28 late orders; 20.68 days late on average

Response: separate volume management, high-rate investigation, and long-delay review.

Cross-state delivery is riskier

Cross-state



Within state



Cross-state orders are 92.22% of RJ orders and 96.99% of its late orders. Mean logistics time: 12.67 versus 3.79 days.

58.26%

of RJ late orders missed the estimate by more than 7 days; 33.71% missed by at least 15 days.

Late orders correlate with worse reviews

Proposed actions only | No implementation or outcome has been verified

Low-score rate / reviewed orders with a 1-2 star rating



Mean rating: 4.29 on time / 2.27 late

The low-score gap is 53.16 percentage points. Products and customer support may also affect reviews.

Analytical next step

Align order-level and date definitions. Carrier and tracking-node data could help test the bottleneck.

01 Monitor key geographies

Track volume, late rate, logistics time, and severe delays by state, city, and seller origin.

02 Escalate long delays

Introduce tiered alerts and customer contact, particularly at 15 and 30 days of delay.

03 Calibrate delivery promises

Validate actual route capacity before changing estimates; avoid merely relaxing the promise.

Which seafood-market services merit a pilot?

Market research project | Completed March 2026

Context

Consumer survey and one month of sales records inform priorities for transaction transparency, product trust, and pilot evaluation.

01 What do shoppers need?

Pain points and service appeal

Identify whether demand clusters around one issue or a service bundle.

02 Who should we test with?

Digital and trust perceptions

Use the original three-group competition model to compare needs.

03 What changed in sales?

Daily sales and products

Identify commercial questions requiring further investigation.

402

valid survey responses

31

sales days

9

products recorded

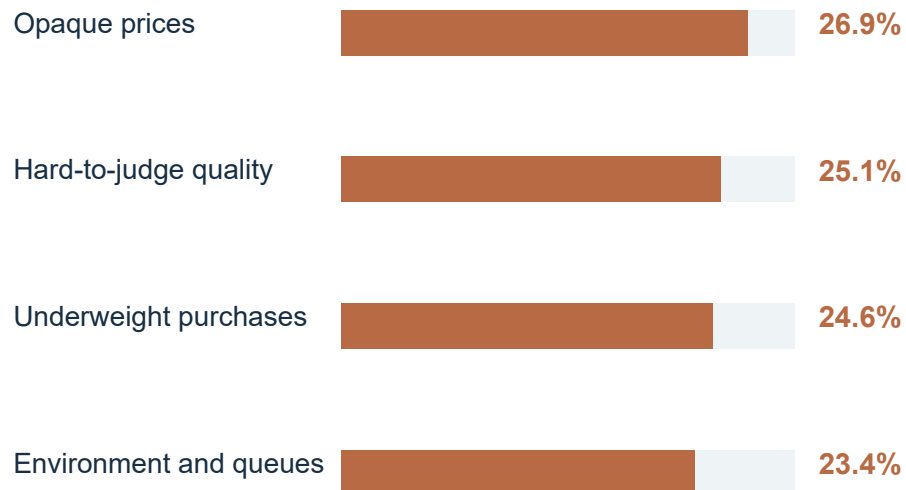
My contribution

Survey design, field research, and core data analysis

Pain points are dispersed; pilot a service bundle

402 survey responses | Share selecting a pain point or agreeing with a feature

Primary pain point / share of responses

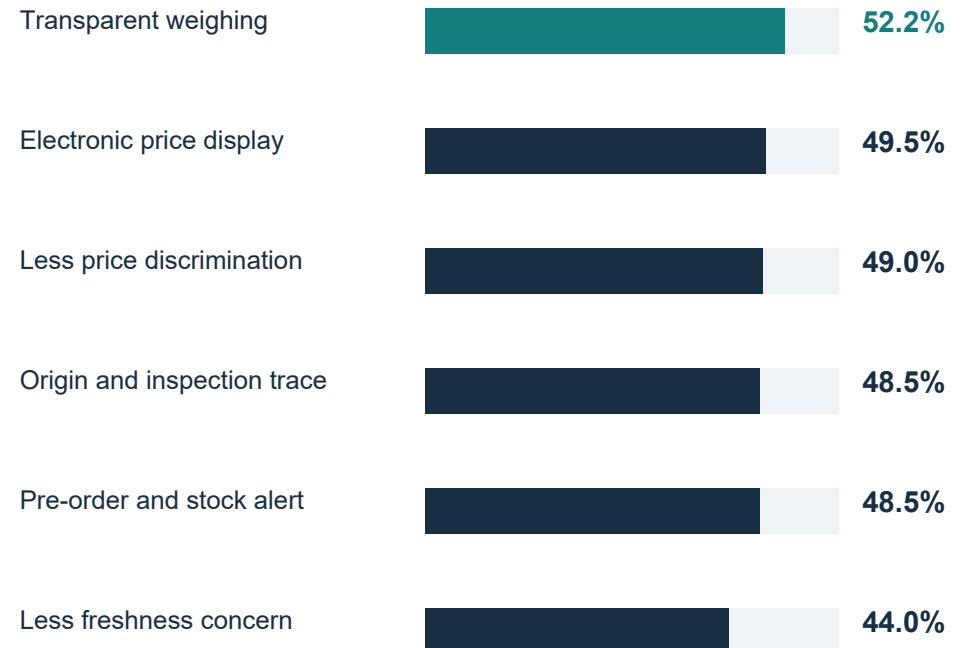


The four shares differ by only 3.5 percentage points; no single issue dominates.

Business implication

Pilot weighing, visible pricing, and verifiable traceability together. Record use, complaints, and repeat visits before scaling.

Support for digital services



Three groups differ in digital and trust needs

K-means | k=3 | Cluster inputs: digital perception S and trust-related concern O

High S / high O

36.07%

Digital perception S	4.09
Trust concern O	4.46
Behavioral intent R	3.74

Higher S / lower O

27.86%

Digital perception S	3.88
Trust concern O	2.60
Behavioral intent R	3.21

Lower S / mid O

36.07%

Digital perception S	2.31
Trust concern O	2.90
Behavioral intent R	2.94

Pilot implications

Invite the high-S/high-O group to trial weighing and traceability. Test convenience with higher-S shoppers and visible in-market tools with lower-S shoppers.

Additional regression check

S and O each correlate positively with intent. With demographic and purchase controls, coefficients are +0.235 and +0.272.

Difference test p=0.645: no evidence that the trust association is stronger.

Interpretation limit

Clustering excludes purchase frequency and observed conversion. R is stated intent, not realized purchases or revenue.

July revenue fell alongside sales volume

July 2025 | 31 days | 9 products | 248 product-day records

CNY 6.238m

Monthly sales

CNY 201k

Daily average

-29.3%

Last 7 vs first 7 days

-27.4%

Volume over same periods

Pattern

Realized sales per jin fell only 2.6%, while volume fell 27.4%. Giant freshwater prawns contributed 24.9% of sales; the top five products, 73.3%.

Recommended validation order

01 Test transaction transparency

Trial weighing, digital price tags, and trace codes; track use, complaints, and repeat visits.

02 Tailor customer guidance

Start with transparent tools for the high-S/high-O group; test alternatives by segment.

03 Diagnose volume decline

Add footfall, stockout, inventory, and loss records for core products.

Data limit

One month cannot establish seasonality. Verified purchase and logistics costs are unavailable, so realized margins are not inferred.

How can health-system capacity be measured?

Chongqing: 38 districts and counties | 2020-2024 panel | Competition project

Context

Connect demographic change with care resources to inform capacity reserves, primary care, and regional coordination.

01 Measure resilience

Resources, staff, access, load

Create a comparable four-dimension proxy index.

02 Examine association

Aging, economy, population

Use lagged variables and panel models to examine conditional patterns.

03 Inform planning

Index and underlying metrics

Translate evidence into monitoring and allocation proposals.

38

districts and counties

190

county-year observations

152

lagged-model observations

My contribution

Data collection, analysis, model development, and conclusion writing

Four dimensions of capacity and load

Entropy weighting combines direction-aligned, normalized indicators

Dimension and proxy metric

Original entropy weight



From source data to index

Direction and units > Global normalization > Entropy weights > County-year index

Weights capture variation in this sample; they are not causal importance or a budget allocation formula.

The aging association varies with the economy

Year fixed effects | District-clustered standard errors | N=152

Without interaction

+0.0142

Coefficient on lagged aging rate

Positive conditional association in this model
95% CI [0.0029, 0.0255]

With economic interaction

Lagged aging rate -0.1875

Aging x log income +0.0168

Both $p < 0.01$; interpret coefficients jointly.

Marginal association

$$-0.1875 + 0.0168 \times \ln(\text{GDP per capita})$$

Interpretation limit

Observed log-GDP range: about 10.24-12.59. Point estimates span -0.0155 to +0.0240; 11.16 is a zero-crossing estimate, not a policy threshold. Year effects retain cross-county differences; causal identification is not claimed.

Alternative proxy aligns; regional results need care

Old-age dependency-ratio check | Subgroup regressions with interaction

Alternative aging proxy / N=152

Lagged dependency ratio -12.287

Ratio x economic level +1.0975

Both $p < 0.01$; directions match the baseline interaction model.
The main effect still depends on economic level.

Subgroup models / saved code output

Core districts: 36

Other districts: 116

Aging term -0.0893*

Aging term -0.0073

Interaction +0.0086**

Interaction +0.0008

* $p < 0.05$; ** $p < 0.01$ within groups. No between-group test.

Operational implication

The alternative measure supports attention to economic context. Subgroup estimates suggest that regional monitoring may need different starting points.

Method limit

The subgroup model omits the GDP main effect and lacks a formal coefficient-difference test. Stronger impact in core districts is not established.

Turn evidence into monitoring and planning

Recommendations only | No implementation or outcome evaluation

Plan ahead

Question: demographic change and resource planning have a time dimension.

Proposal: jointly track population, beds, staff, and operating load for earlier preparation.

Limit: alert thresholds and allocation quantities have not been field-tested.

Build primary-care capacity

Question: the composite index does not replace underlying service metrics.

Proposal: expand capacity for chronic disease, rehabilitation, and long-term care.

Limit: index weights should not be converted directly into funding shares.

Coordinate across regions

Question: care capacity differs between core and non-core districts.

Proposal: improve collaboration and referrals between districts and core-city hospitals.

Limit: patient-flow mechanisms and compensation require direct flow data.